```
                HH        HH   AAAAAAAAAA   NN        NN  OOOOOOOOOOOO   IIIIIIIIII   PPPPPPPPPPP   LL             IIIIIIIIII
               HH        HH  AAAAAAAAAAAA  NNN       NN  OOOOOOOOOOOO   IIIIIIIIII   PPPPPPPPPPPP  LL             IIIIIIIIII
              HH        HH  AA        AA  NNNN      NN  OO        OO       II       PP        PP  LL               II
             HH        HH  AA        AA  NN NN     NN  OO        OO       II       PP        PP  LL               II
            HH        HH  AA        AA  NN  NN    NN  OO        OO       II       PP        PP  LL               II
           HHHHHHHHHHHH  AAAAAAAAAAAA  NN   NN   NN  OO        OO       II       PPPPPPPPPPP  LL               II
          HHHHHHHHHHHH  AAAAAAAAAAAA  NN    NN  NN  OO        OO       II       PPPPPPPPPP   LL               II
         HH        HH  AA        AA  NN     NN NN  OO        OO       II       PP           LL               II
        HH        HH  AA        AA  NN      NNNN  OO        OO       II       PP           LL               II
       HH        HH  AA        AA  NN       NNN  OO        OO       II       PP           LL               II
      HH        HH  AA        AA  NN        NN  OOOOOOOOOOOO   IIIIIIIIII   PP           LLLLLLLLLLL   IIIIIIIIII
     HH        HH  AA        AA  NN         N  OOOOOOOOOOOO   IIIIIIIIII   PP           LLLLLLLLLLL   IIIIIIIIII


          JJJJJJJJJJ   9999999999   5555555555   2222222222                          AAAAAAAAAA
          JJJJJJJJJJ  9999999999  5555555555  2222222222                          AAAAAAAAAAAA
              JJ      99        99  55          22        22                        AA        AA
              JJ      99        99  55                    22                        AA        AA
              JJ      99        99  55                    22                        AA        AA
              JJ      9999999999  555555555            22                          AAAAAAAAAAAA
              JJ      9999999999  5555555555         22                            AAAAAAAAAAAA
              JJ              99          55       22                              AA        AA
            JJ    JJ          99           55    22                                AA        AA
            JJ    JJ      99        99        55   22                              AA        AA
            JJJJJJJ      9999999999  5555555555  2222222222                        AA        AA
             JJJJJJ       9999999999  5555555555  2222222222                        AA        AA


****A  START  JOB  952  HANOIPLI  Hanoi rec PL/1       ROOM       8.47.52 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  952  START  A****
****A  START  JOB  952  HANOIPLI  Hanoi rec PL/1       ROOM       8.47.52 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  952  START  A****
****A  START  JOB  952  HANOIPLI  Hanoi rec PL/1       ROOM       8.47.52 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  952  START  A****
```

J E S 2 J O B L O G

```
20.47.51 JOB  952  $HASP373 HANOIPLI STARTED - INIT  1 - CLASS A - SYS TK4-
20.47.51 JOB  952  IEF403I HANOIPLI - STARTED - TIME=20.47.51
20.47.52 JOB  952  IEFACTRT - Stepname  Procstep  Program   Retcode
20.47.52 JOB  952  HANOIPLI   HANOI     PL1L      IEMAA     RC= 0000
20.47.52 JOB  952  HANOIPLI   HANOI     GO        LOADER    RC= 0000
20.47.52 JOB  952  IEF404I HANOIPLI - ENDED - TIME=20.47.52
20.47.52 JOB  952  $HASP395 HANOIPLI ENDED


------ JES2 JOB STATISTICS ------

 01 NOV 26 JOB EXECUTION DATE

       105 CARDS READ

       313 SYSOUT PRINT RECORDS

         0 SYSOUT PUNCH RECORDS

      0.02 MINUTES EXECUTION TIME
```

```
 1      //HANOIPLI JOB (MBA),                                                JOB  952
        //             'Hanoi rec PL/1',
        //             CLASS=A,
        //             MSGCLASS=A,
        //             NOTIFY=HERC01,
        //             REGION=8M,TIME=1440,
        //             MSGLEVEL=(1,1),
        //           USER=HERC01,PASSWORD=           GENERATED BY GDL
        ************************************************************************
        *** Name: HERC01.MYSTUFF.JCL(HANOIPLI)
        ***
        *** Desc: Towers of Hanoi.
        ***       Number of slices n (1<=n<=10), source stack s (0<=s<=2)
        ***       and destination stack d (0<=d<=2)&(d<>s)) are entered
        ***       in //GO.SYSIN DD
        ************************************************************************
 2      //HANOI    EXEC PL1LFCG,PARM.PL1L='LOAD,NODECK,OPT=02'
 3      XXPL1LFCG PROC SOUT='*'                                                 00000100
 4      XXPL1L   EXEC  PGM=IEMAA,PARM='LOAD,NODECK'                             00000200
 5      XXSYSPRINT DD  SYSOUT=&SOUT                                             00000300
 6      //PL1L.SYSLIN DD UNIT=SYSDA
        X/SYSLIN   DD  DSNAME=&&LOADSET,DISP=(MOD,PASS),UNIT=SYSSQ,            *00000400
        XX             SPACE=(80,(250,100))                                     00000500
 7      XXSYSUT3  DD  DSNAME=&&SYSUT3,UNIT=SYSDA,SPACE=(80,(250,250)),         *00000600
        XX             DCB=BLKSIZE=80                                           00000700
 8      XXSYSUT1   DD  DSNAME=&&SYSUT1,UNIT=SYSDA,SPACE=(1024,(60,60),,CONTIG),*00000800
        XX             SEP=(SYSUT3,SYSLIN),DCB=BLKSIZE=1024                     00000900
 9      //PL1L.SYSIN DD *
10      XXGO     EXEC  PGM=LOADER,PARM='MAP,PRINT',                           *00001000
        XX             COND=(9,LT,PL1L)                                         00001100
11      XXSYSLIB   DD  DSNAME=SYS1.PL1LIB,DISP=SHR                              00001200
12      XXSYSLIN   DD  DSNAME=&&LOADSET,DISP=(OLD,DELETE)                       00001300
13      XXSYSLOUT  DD  SYSOUT=&SOUT                                             00001400
14      //GO.SYSPRINT DD SYSOUT=*,DCB=(RECFM=FBA,LRECL=161,BLKSIZE=16100)
        X/SYSPRINT DD  SYSOUT=&SOUT                                             00001500
15      //GO.SYSIN DD *
```

```
 STMT NO. MESSAGE
-
    5     IEF653I SUBSTITUTION JCL - SYSOUT=*
   13     IEF653I SUBSTITUTION JCL - SYSOUT=*
   14     IEF653I SUBSTITUTION JCL - SYSOUT=*
IEF236I ALLOC. FOR HANOIPLI PL1L HANOI
IEF237I JES2 ALLOCATED TO SYSPRINT
IEF237I 180  ALLOCATED TO SYSLIN
IEF237I 170  ALLOCATED TO SYSUT3
IEF237I 140  ALLOCATED TO SYSUT1
IEF237I JES2 ALLOCATED TO SYSIN
IEF142I HANOIPLI PL1L HANOI - STEP WAS EXECUTED - COND CODE 0000
IEF285I   JES2.JOB00952.SO0103                          SYSOUT
IEF285I   SYS26305.T204751.RA000.HANOIPLI.LOADSET       PASSED        *-------73
IEF285I   VOL SER NOS= WORK02.
IEF285I   SYS26305.T204751.RA000.HANOIPLI.SYSUT3        DELETED       *--------0
IEF285I   VOL SER NOS= WORK01.
IEF285I   SYS26305.T204751.RA000.HANOIPLI.SYSUT1        DELETED       *--------0
IEF285I   VOL SER NOS= WORK00.
IEF285I   JES2.JOB00952.SI0101                          SYSIN
IEF373I STEP /PL1L    / START 26305.2047
IEF374I STEP /PL1L    / STOP  26305.2047 CPU    0MIN 00.44SEC SRB    0MIN 00.15SEC VIRT  8192K SYS   228K
********************************************************************************************************************
*     1. Jobstep of job: HANOIPLI    Stepname: PL1L        Program name: IEMAA      Executed on 01.11.26 from 20.47.51 to 20.47.52 *
*         elapsed time  00:00:00,98                    CPU-Identifier:  TK4-          Page-in:      0                          *
*             CPU time  00:00:00,59              Virtual Storage used:   8192K         Page-out:     0                          *
*           corr. CPU:  00:00:00,59   CPU time has been corrected by  1 / 1,0  multiplier                                      *
*                                                                                                                              *
*     I/O Operation                                                                                                            *
*     Number of records read via DD * or DD DATA:     81                                                                       *
*     DMY.......0 180......73 170.......0 140.......0 DMY.......0                                                              *
*                                                                                                                              *
*                                      Charge for step (w/o SYSOUT):          0,98                                         *
********************************************************************************************************************
IEF236I ALLOC. FOR HANOIPLI GO HANOI
IEF237I 148  ALLOCATED TO SYSLIB
IEF237I 180  ALLOCATED TO SYSLIN
IEF237I JES2 ALLOCATED TO SYSLOUT
IEF237I JES2 ALLOCATED TO SYSPRINT
IEF237I JES2 ALLOCATED TO SYSIN
IEF142I HANOIPLI GO HANOI - STEP WAS EXECUTED - COND CODE 0000
IEF285I   SYS1.PL1LIB                                   KEPT          *------155
IEF285I   VOL SER NOS= MVSRES.
IEF285I   SYS26305.T204751.RA000.HANOIPLI.LOADSET       DELETED       *-------74
IEF285I   VOL SER NOS= WORK02.
IEF285I   JES2.JOB00952.SO0104                          SYSOUT
IEF285I   JES2.JOB00952.SO0105                          SYSOUT
IEF285I   JES2.JOB00952.SI0102                          SYSIN
IEF373I STEP /GO      / START 26305.2047
IEF374I STEP /GO      / STOP  26305.2047 CPU    0MIN 00.14SEC SRB    0MIN 00.05SEC VIRT   332K SYS   212K
********************************************************************************************************************
*     2. Jobstep of job: HANOIPLI    Stepname: GO          Program name: LOADER     Executed on 01.11.26 from 20.47.52 to 20.47.52 *
*         elapsed time  00:00:00,24                    CPU-Identifier:  TK4-          Page-in:      0                          *
*             CPU time  00:00:00,19              Virtual Storage used:    332K         Page-out:     0                          *
*           corr. CPU:  00:00:00,19   CPU time has been corrected by  1 / 1,0  multiplier                                      *
*                                                                                                                              *
*     I/O Operation                                                                                                            *
*     Number of records read via DD * or DD DATA:     1                                                                        *
*     148.....155 180......74 DMY.......0 DMY.......0 DMY.......0                                                              *
*                                                                                                                              *
*                                      Charge for step (w/o SYSOUT):          0,31                                         *
********************************************************************************************************************
IEF375I  JOB /HANOIPLI/ START 26305.2047
```

```
IEF376I  JOB /HANOIPLI/ STOP  26305.2047 CPU    0MIN 00.58SEC SRB    0MIN 00.20SEC
```

PL/I F COMPILER OPTIONS SPECIFIED ARE AS FOLLOWS--

```
     LOAD,NODECK,OPT=02


THE COMPLETE LIST OF OPTIONS USED DURING THIS COMPILATION IS--   EBCDIC
                                                                CHAR60
                                                              NOMACRO
                                                                SOURCE2
                                                              NOMACDCK
                                                                COMP
                                                                SOURCE
                                                              NOATR
                                                              NOXREF
                                                              NOEXTREF
                                                              NOLIST
                                                                LOAD
                                                              NODECK
                                                                FLAGW
                                                              NOSTMT
                                                                SIZE=8351184
                                                                LINECNT=050
                                                                OPT=02
                                                                SORMGIN=(002,072)
                                                              NOEXTDIC
                                                              NONEST
                                                              OPLIST
                                                              SYNCHKT

*OPTIONS IN EFFECT*     EBCDIC,CHAR60,NOMACRO,SOURCE2,NOMACDCK,COMP,SOURCE,NOATR,NOXREF,NOEXTREF,NOLIST,LOAD,
*OPTIONS IN EFFECT*     NODECK,FLAGW,NOSTMT,SIZE=8351184,LINECNT=050,OPT=02,SORMGIN=(002,072),NOEXTDIC,
*OPTIONS IN EFFECT*     NONEST,OPLIST,SYNCHKT
```

```
 1              HANOI: PROC OPTIONS (MAIN) REORDER;
                /* ////////////////////////////////////////////////////////////
                 * Name: ManInBlueAT
                 * Program: The Towers of Hanoi
                 * Language: PL/1
                 *
                 * Desc:
                 * Move n slices from source stack s to destination stack d using help
                 * stack h (3-s-d). A recursive algorithm is applied. Please refer to
                 * https://en.wikipedia.org/wiki/Tower_of_Hanoi
                 *
                 * Changes:
                 * - ManInBlueAT, 2024-05-20 New program
                 * //////////////////////////////////////////////////////////// */
 2              DECLARE N      BINARY FIXED(31);
 3              DECLARE S      BINARY FIXED(31);
 4              DECLARE D      BINARY FIXED(31);
 5              DECLARE H      BINARY FIXED(31);
 6              DECLARE STEPS  BINARY FIXED(31);

 7              DECLARE SYSIN FILE;
 8              DECLARE SYSPRINT FILE;

 9              GET LIST(N,S,D) FILE(SYSIN);
10              H = 3 - S - D;

11              PUT EDIT('Moving ',N,' slices from source stack ',S,
                  ' to destination stack ',D,' using help stack ',H)
                  (A(7),F(2,0),A(26),F(2,0),A(22),F(2,0),A(18),F(2,0))
                  FILE(SYSPRINT);
12              PUT SKIP FILE(SYSPRINT);

13              STEPS = 0;
14              STEPS = DOHANOI(N,S,D,H,STEPS);

15              PUT SKIP(2) EDIT('Done.')(A(5)) FILE(SYSPRINT);

                /* ------------------------------------------------------------------
                 * Recursive function DOHANOI that processes the sequence of steps:
                 * Move uppper n-1 slices from source to help stack, move lowest slice
                 * (slice n) from source to destination stack and move upper n-1 slices
                 * from help stack to destination stack
                 * ------------------------------------------------------------------ */
16              DOHANOI: PROCEDURE(N,S,D,H,STEPS) RECURSIVE;
17                DECLARE N      BINARY FIXED(31);
18                DECLARE S      BINARY FIXED(31);
19                DECLARE D      BINARY FIXED(31);
```

```
20                DECLARE H      BINARY FIXED(31);
21                DECLARE STEPS  BINARY FIXED(31);

22                IF (N > 1) THEN DO;
24                  STEPS = DOHANOI(N-1,S,H,D,STEPS);
25                  STEPS = DOSTEP(N,S,D,STEPS);
26                  STEPS = DOHANOI(N-1,H,D,S,STEPS);
27                  END;
28                ELSE
28                  STEPS = DOSTEP(N,S,D,STEPS);

29                RETURN(STEPS);
30              END DOHANOI;

                /* ------------------------------------------------------------------
                 * DOSTEP performs the step of moving a slice. The counter STEPS is
                 * increased and the current slice i is moved from the current source
                 * stack sc to the current destination stack dc
                 * ------------------------------------------------------------------ */
31              DOSTEP: PROCEDURE(I,SC,DC,STEPS);
32                DECLARE I      BINARY FIXED(31);
33                DECLARE SC     BINARY FIXED(31);
34                DECLARE DC     BINARY FIXED(31);
35                DECLARE STEPS  BINARY FIXED(31);
36                STEPS = STEPS + 1;
37                PUT SKIP EDIT('Step ',STEPS,': Moving slice ',I,' from stack ',
                    SC,' to stack ',DC)
                    (A(5),F(5,0),A(15),F(2,0),A(12),F(2,0),A(10),F(2,0))
                    FILE(SYSPRINT);

38                RETURN(STEPS);
39              END DOSTEP;

40              END HANOI;
```

STORAGE REQUIREMENTS.

THE STORAGE AREA FOR THE PROCEDURE LABELLED HANOI IS 224 BYTES LONG.

THE STORAGE AREA FOR THE PROCEDURE LABELLED DOHANOI IS 232 BYTES LONG.

THE STORAGE AREA (IN STATIC) FOR THE PROCEDURE LABELLED DOSTEP IS 184 BYTES LONG.

THE PROGRAM CSECT IS NAMED HANOI AND IS 1530 BYTES LONG.

THE STATIC CSECT IS NAMED **HANOIA AND IS 648 BYTES LONG.

STATISTICS SOURCE RECORDS = 81,PROG TEXT STMNTS = 40,OBJECT BYTES = 1530

TABLE OF OFFSETS AND STATEMENT NUMBERS WITHIN PROCEDURE DOHANOI

OFFSET (HEX) 0000 0058 0068 0068 00D2 0128 0196 019A 01F0 021A
STATEMENT NO   16   22   23   24   25   26   27   28   29   30

TABLE OF OFFSETS AND STATEMENT NUMBERS WITHIN PROCEDURE DOSTEP

OFFSET (HEX) 0000 0048 0058 0134 015E
STATEMENT NO   31   36   37   38   39

TABLE OF OFFSETS AND STATEMENT NUMBERS WITHIN PROCEDURE HANOI

OFFSET (HEX) 0000 0038 0072 0082 014E 015E 0164 01BE 01F0
STATEMENT NO   1   9   10   11   12   13   14   15   40

 NO ERRORS OR WARNINGS DETECTED.

AUXILIARY STORAGE WILL NOT BE USED FOR DICTIONARY WHEN SIZE = 120K

COMPILE TIME .00 MINS

ELAPSED TIME .01 MINS

VS LOADER

OPTIONS USED - PRINT,MAP,NOLET,CALL,RES,NOTERM,SIZE=307200,NAME=**GO

NAME	TYPE	ADDR	NAME	TYPE	ADDR	NAME	TYPE	ADDR	NAME	TYPE	ADDR	NAME	TYPE	ADDR
HANOI	SD	AC010	**HANOIA	SD	AC610	IHEMAIN	SD	AC898	IHENTRY	SD	AC8A0	IHESPRT	SD	AC8B0
SYSIN	SD	AC8E8	IHESAP *	SD	AC920	IHESADA *	LR	AC920	IHESAPC *	LR	AC93A	IHESAPD *	LR	AC942
IHESAPA *	LR	AC94A	IHESAPB *	LR	AC952	IHESADF *	LR	AC95A	IHESADB *	LR	AC962	IHESADE *	LR	AC96A
IHESAFC *	LR	AC972	IHESAFA *	LR	AC97A	IHESAFB *	LR	AC982	IHESAFD *	LR	AC98A	IHESARA *	LR	AC992
IHESAFQ *	LR	AC99A	IHESARC *	LR	AD18C	IHESADD *	LR	AD2A0	IHESAFF *	LR	AD2DA	IHEBEG *	SD	AD440
IHEBEGN *	LR	AD440	IHEBEGA *	LR	AD480	IHEDDO *	SD	AD4C0	IHEDDOA *	LR	AD4C0	IHEDDOB *	LR	AD4C2
IHEDDOC *	LR	AD4C4	IHEDDOD *	LR	AD4C6	IHEDDOE *	LR	AD4C8	IHEERR *	SD	AD748	IHEERRD *	LR	AD748
IHEERRC *	LR	AD752	IHEERRB *	LR	AD75C	IHEERRA *	LR	AD766	IHEERRE *	LR	ADDF6	IHEOCL *	SD	ADE90
IHEOCLA *	LR	ADE90	IHEOCLB *	LR	ADE92	IHEOCLC *	LR	ADE94	IHEOCLD *	LR	ADE96	IHEPRT *	SD	AE410
IHEPRTA *	LR	AE410	IHEPRTB *	LR	AE412	IHESIZ *	SD	AE718	IHESIZE *	LR	AE718	IHETAB *	SD	AE728
IHETABS *	LR	AE728	IHEDDP *	SD	AE738	IHEDDPA *	LR	AE738	IHEDDPB *	LR	AE73A	IHEDDPC *	LR	AE73C
IHEDDPD *	LR	AE73E	IHEIOF *	SD	AE9C8	IHEIOFB *	LR	AE9C8	IHEIOFA *	LR	AE9CA	IHEITAZ *	LR	AEC66
IHEITAX *	LR	AEC72	IHEITAA *	LR	AEC86	IHELDO *	SD	AECA8	IHELDOA *	LR	AECA8	IHELDOB *	LR	AECAA
IHELDOC *	LR	AECAE	IHEDNC *	SD	AF0C0	IHEDNCA *	LR	AF0C0	IHEVSB *	SD	AF378	IHEVSBA *	LR	AF378
IHEVSC *	SD	AF448	IHEVSCA *	LR	AF448	IHEABN *	SD	AF4F8	IHEABND *	LR	AF4F8	IHEM91 *	SD	AF508
IHEM91B *	LR	AF508	IHEM91A *	LR	AF510	IHEM91C *	LR	AF622	IHETER *	SD	AF680	IHETERA *	LR	AF680
IHEDMA *	SD	AF790	IHEDMAA *	LR	AF790	IHEUPA *	SD	AF888	IHEUPAA *	LR	AF888	IHEUPAB *	LR	AF8F2
IHEVQC *	SD	AF970	IHEVQCA *	LR	AF970	IHEVSE *	SD	AFBD8	IHEVSEA *	LR	AFBD8	IHEVSEB *	LR	AFBDA
IHEVSF *	SD	AFD38	IHEVSFA *	LR	AFD38	IHEVFA *	SD	AFE28	IHEVFAA *	LR	AFE28	IHEVFB *	SD	AFF98
IHEVFBA *	LR	AFF98	IHEVFC *	SD	B0088	IHEVFCA *	LR	B0088	IHEVFD *	SD	B00B0	IHEVFDA *	LR	B00B0
IHEVFE *	SD	B0118	IHEVFEA *	LR	B0118	IHEVKB *	SD	B0138	IHEVKBA *	LR	B0138	IHEVKC *	SD	B0448
IHEVKCA *	LR	B0448	IHEVKF *	SD	B0718	IHEVKFA *	LR	B0718	IHEVKG *	SD	B0D98	IHEVKGA *	LR	B0D98
IHEVPA *	SD	B1278	IHEVPAA *	LR	B1278	IHEVPB *	SD	B1458	IHEVPBA *	LR	B1458	IHEVPC *	SD	B1600
IHEVPCA *	LR	B1600	IHEVPD *	SD	B17F0	IHEVPDA *	LR	B17F0	IHEVPE *	SD	B18F8	IHEVPEA *	LR	B18F8
IHEVPF *	SD	B1B70	IHEVPFA *	LR	B1B70	IHEVPG *	SD	B1BC0	IHEVPGA *	LR	B1BC0	IHEVPH *	SD	B1DF0
IHEVPHA *	LR	B1DF0	IHELDI *	SD	B1EA8	IHELDIA *	LR	B1EA8	IHELDIB *	LR	B1EAA	IHELDIC *	LR	B1EAC
IHELDID *	LR	B1EAE	IHEDCN *	SD	B2720	IHEDCNA *	LR	B2720	IHEDCNB *	LR	B2722	IHEDNB *	SD	B2978
IHEDNBA *	LR	B2978	IHEKCD *	SD	B2A90	IHEKCDA *	LR	B2A90	IHEKCDB *	LR	B2A92	IHEUPB *	SD	B2BA0
IHEUPBA *	LR	B2BA0	IHEUPBB *	LR	B2C08	IHEVCA *	SD	B2C88	IHEVCAA *	LR	B2C88	IHEVCS *	SD	B2DA0
IHEVCSA *	LR	B2DA0	IHEVCSB *	LR	B2DA2	IHEVSD *	SD	B2F80	IHEVSDA *	LR	B2F80	IHEVSDB *	LR	B2F82
IHEVQB *	SD	B3120	IHEVQBA *	LR	B3120	IHEVQA *	SD	B3600	IHEVQAA *	LR	B3600	IHEVSA *	SD	B3700
IHEVSAA *	LR	B3700	IHEDOB *	SD	B3840	IHEDOBA *	LR	B3840	IHEDOBB *	LR	B3842	IHEDOBC *	LR	B3844
IHEIOD *	SD	B3988	IHEIODG *	LR	B3988	IHEIODP *	LR	B398A	IHEIODT *	LR	B3A82	IHEDOA *	SD	B3C28
IHEDOAA *	LR	B3C28	IHEDOAB *	LR	B3C2A	IHEDBN *	SD	B3E68	IHEDBNA *	LR	B3E68	IHEIOB *	SD	B3FD0
IHEIOBA *	LR	B3FD0	IHEIOBB *	LR	B3FD8	IHEIOBC *	LR	B3FE0	IHEIOBD *	LR	B3FE8	IHEIOBE *	LR	B3FF0
IHEIOBT *	LR	B40D8	IHEIOP *	SD	B41B0	IHEIOPA *	LR	B41B0	IHEIOPB *	LR	B41B2	IHEIOPC *	LR	B41B6
IHEIOA *	SD	B43A8	IHEIOAA *	LR	B43A8	IHEIOAB *	LR	B43AA	IHEIOAC *	LR	B43AC	IHEIOAD *	LR	B43AE
IHEIOAT *	LR	B4498												

IHEQINV	PR	00	IHEQERR	PR	4	IHEQTIC	PR	8	IHEQLWF	PR	C	IHEQSLA	PR	10
IHEQLW0	PR	14	**HANOIB	PR	18	**HANOIC	PR	1C	**HANOID	PR	20	IHEQSPR	PR	24
SYSIN	PR	28	IHEQLSA	PR	2C	IHEQLW1	PR	30	IHEQLW2	PR	34	IHEQLW3	PR	38
IHEQLW4	PR	3C	IHEQLWE	PR	40	IHEQLCA	PR	44	IHEQVDA	PR	48	IHEQFVD	PR	4C

```
IHEQFOP    PR     50    IHEQCFL    PR     54    IHEQADC    PR     5C    IHEQLPR    PR     60    IHEQSAR    PR     64
IHEQRTC    PR     68    IHEQSFC    PR     6C    IHEQXLV    PR     70    IHEQEVT    PR     78

TOTAL LENGTH     8508
ENTRY ADDRESS   AC8A0
```

```
Moving  4 slices from source stack  0 to destination stack  1 using help stack  2

Step     1: Moving slice  1 from stack  0 to stack  2
Step     2: Moving slice  2 from stack  0 to stack  1
Step     3: Moving slice  1 from stack  2 to stack  1
Step     4: Moving slice  3 from stack  0 to stack  2
Step     5: Moving slice  1 from stack  1 to stack  0
Step     6: Moving slice  2 from stack  1 to stack  2
Step     7: Moving slice  1 from stack  0 to stack  2
Step     8: Moving slice  4 from stack  0 to stack  1
Step     9: Moving slice  1 from stack  2 to stack  1
Step    10: Moving slice  2 from stack  2 to stack  0
Step    11: Moving slice  1 from stack  1 to stack  0
Step    12: Moving slice  3 from stack  2 to stack  1
Step    13: Moving slice  1 from stack  0 to stack  2
Step    14: Moving slice  2 from stack  0 to stack  1
Step    15: Moving slice  1 from stack  2 to stack  1

Done.
```

```
                HH        HH   AAAAAAAAAA   NN        NN  OOOOOOOOOOOO   IIIIIIIIII   PPPPPPPPPPP   LL             IIIIIIIIII
               HH        HH  AAAAAAAAAAAA  NNN       NN  OOOOOOOOOOOO   IIIIIIIIII   PPPPPPPPPPPP  LL             IIIIIIIIII
              HH        HH  AA        AA  NNNN      NN  OO        OO       II       PP        PP  LL                II
             HH        HH  AA        AA  NN NN     NN  OO        OO       II       PP        PP  LL                II
            HH        HH  AA        AA  NN  NN    NN  OO        OO       II       PP        PP  LL                II
           HHHHHHHHHHHH  AAAAAAAAAAAA  NN   NN   NN  OO        OO       II       PPPPPPPPPPPP  LL                II
          HHHHHHHHHHHH  AAAAAAAAAAAA  NN    NN  NN  OO        OO       II       PPPPPPPPPPP   LL                II
         HH        HH  AA        AA  NN     NN NN  OO        OO       II       PP            LL                II
        HH        HH  AA        AA  NN      NNNN  OO        OO       II       PP            LL                II
       HH        HH  AA        AA  NN       NNN  OO        OO       II       PP            LL                II
      HH        HH  AA        AA  NN        NN  OOOOOOOOOOOO   IIIIIIIIII   PP            LLLLLLLLLLL   IIIIIIIIII
     HH        HH  AA        AA  NN         N  OOOOOOOOOOOO   IIIIIIIIII   PP            LLLLLLLLLLL   IIIIIIIIII


                JJJJJJJJJJ   9999999999   5555555555   2222222222                          AAAAAAAAAA
                JJJJJJJJJJ  9999999999  5555555555  2222222222                          AAAAAAAAAAAA
                    JJ      99        99  55          22        22                        AA        AA
                    JJ      99        99  55                    22                        AA        AA
                    JJ      99        99  55                    22                        AA        AA
                    JJ      9999999999  555555555            22                          AAAAAAAAAAAA
                    JJ      9999999999  5555555555         22                            AAAAAAAAAAAA
                    JJ              99          55       22                              AA        AA
                JJ  JJ              99           55    22                                AA        AA
                JJ  JJ      99      99           55  22                                  AA        AA
                JJJJJJJ     9999999999  5555555555  2222222222                          AA        AA
                 JJJJJ       9999999999   55555555   2222222222                          AA        AA


****A   END   JOB  952  HANOIPLI  Hanoi rec PL/1       ROOM       8.47.52 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  952   END   A****
****A   END   JOB  952  HANOIPLI  Hanoi rec PL/1       ROOM       8.47.52 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  952   END   A****
****A   END   JOB  952  HANOIPLI  Hanoi rec PL/1       ROOM       8.47.52 PM 01 NOV 26  PRINTER1  SYS TK4-  JOB  952   END   A****
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